

MECHANISM OF ACTION OF NON-LYTIC ANTIMICROBIAL PEPTIDES STUDIED BY *IN VIVO* SOLID-STATE NMR AND FLUORESCENCE MICROSCOPY

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ABSTRACT

Non-lytic antimicrobial peptides (AMPs) kill bacteria by reaching the cytoplasm and binding with an intracellular target, but the way they cross the cell membrane is poorly understood [1-2]. We used *in vivo* ²H solid-state NMR to study how the translocation of several non-lytic AMPs, such as buforin and oncocin, affects the order and rigidity of the deuterium-labeled membranes of *E. coli* [3]. We also used timelapse fluorescence microscopy and microfluidics to get insight into the succession of events induced by the presence of AMPs at the membrane level [4]. We find that, even though the effect of these non-lytic AMPs on the membrane should be limited and transient, they actually have quite distinguishable effects on the membrane rigidity and permeability. We also find that those effects vary from one peptide to another, confirming a wide diversity of mechanisms used to cross the membrane.

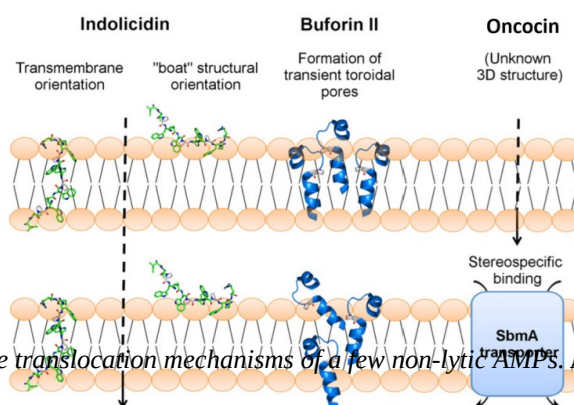


Fig. 1: Membrane translocation mechanisms of a few non-lytic AMPs. Adapted from [1-2].

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